Project Trimester 2 2026

COMP1013 Analytics Programming

Due Friday 24 July 2026 (Week 11)

1 Project Description

In this assignment there are 5 parts. For each of parts 1-4 you should:

- Explain your rationales behind choices made answering the tasks.
- Write the appropriate R code.
- Include comments within the code to explain the algorithm.
- Test the code to ensure its correctness.
- Format and structure the code to maximise its readability.

For part 5 you should use Git and GitHub to manage your development process. Make commits with meaningful messages.

A report must be submitted containing a cover page, the solutions to each of the four parts, and your code, as a **PDF**, to the vUWS submission site. The cover page must contain your name, student number, subject code and name, and the declaration below.

Submissions in other formats or without cover pages will have marks deducted.

Submission is due by Friday of week 13. Late submissions will receive a 10% reduction in marks for each day late.

2 Marking Criteria

This assignment is worth 40% of the subject assessment tasks. There four problems to investigate and 10 marks are available for each of the four problems. In addition, there is 10 marks for using of Git in the assignment. Therefore, the **total marks** for the assignment is **50**. The marking criteria for Questions 1-4 are given in Table 1, while the marking criteria for Question 5 (Git) are presented in Table 2.

Criteria	Q1	Q2	Q3	Q4
Rationales of algorithm choices (1 mark) Code Correctness (4 marks) Comments explaining code (2 marks) Code Testing (1 mark) Code Style and Readability (2 marks)				
Total (10 marks)				

Table 1: Marking criteria for Questions 1-4.

Criteria	Q5
GitHub repository created and link provided in the report (2 mark) Scripts and datasets uploaded with an appropriate folder structure (2 marks) Three commits made during the project development process (2 marks) Clear and meaningful commit messages reflecting project progress (2 mark) Commit history screenshot with a brief explanation of each commit (2 marks)	
Total (10 marks)	

Table 2: Marking criteria for Question 5 (Git).

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When writing the solutions to each of the four parts, make sure to consult the marking criteria and check that you have covered them. The project will be marked using this criteria.

For each task, there are a maximum of 2 bonus marks available for answers above and beyond the subject content. For example, extra analysis.

3 Declaration

Before submitting the assignment, include the following declaration in a clearly visible and readable place on the cover page of your project report.

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By including this statement, we the authors of this work, verify that:

- We hold a copy of this assignment that we can produce if the original is lost or damaged.
- We hereby certify that no part of this assignment/product has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.
- No part of this assignment/product has been written/produced for us by another person except where such collaboration has been authorised by the subject lecturer/tutor concerned.
- We are aware that this work may be reproduced and submitted to plagiarism detection software programs for the purpose of detecting possible plagiarism (**which may retain a copy on its database for future plagiarism checking**).
- We hereby certify that we have read and understand what the School of Computing, Engineering and Mathematics defines as minor and substantial breaches of misconduct as outlined in the learning guide for this unit.

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Note: An examiner or lecturer/tutor has the right not to mark this project report if the above declaration has not been added to the cover of the report.

4 Project Tasks

You are working as a data scientist and analyst, focusing on FIFA World Cup data analysis. Your task is to analyze match data, team performance, player statistics, and tournament-related reports. The dataset is divided into four subsets:

Matches: Match data:

– **MatchID**	Unique identifier for each match.
– **TournamentID**	Identifier of the FIFA World Cup tournament in which the match was played.
– **MatchName**	Official name or description of the match (e.g., "Final", "Argentina vs France").
– **Stage**	Competition stage of the match, such as Group Stage, Round of 16, Quarter-finals, Semi-finals, Third-place Play-off, or Final.
– **MatchDate**	Date on which the match was played.
– **MatchTime**	Kick-off time of the match.
– **StadiumID**	Identifier of the stadium where the match took place.
– **HomeTeamID**	Identifier of the home team.
– **AwayTeamID**	Identifier of the away team.
– **HomeTeamScore**	Number of goals scored by the home team during regular and extra time.
– **AwayTeamScore**	Number of goals scored by the away team during regular and extra time.
– **ExtraTime**	Indicates whether the match proceeded to extra time ("TRUE"/"FALSE" or equivalent).
– **PenaltyShootout**	Indicates whether the match was decided by a penalty shootout.
– **HomePenalty**	Number of penalty goals scored by the home team during the penalty

		shootout.
–	**AwayPenalty**	Number of penalty goals scored by the away team during the penalty shootout.
–	**Result**	Outcome of the match, typically indicating the winning team or whether the match ended in a draw.

Teams: Team data:

–	**TeamID**	Unique identifier for each national team.
–	**TeamName**	Official name of the national team.
–	**TeamCode**	Three-letter FIFA country code representing the team (e.g., BRA for Brazil, ARG for Argentina).

Stadiums: Stadium data:

–	**StadiumID**	Unique identifier for each stadium.
–	**StadiumName**	Official name of the stadium.
–	**City**	City where the stadium is located.
–	**Country**	Country in which the stadium is located.

Tournaments: Tournament data:

–	**TournamentID**	Unique identifier for a FIFA World Cup tournament edition.
–	**TournamentName**	Official name of the tournament (e.g., "2026 FIFA World Cup").

Your tasks are:

1. Write the code to inspect the data structure and present the data: The missing values in the dataset were written as "?", replace any "?" with NA; Convert categorical variables **Result, Stage, Country**, and **Extratime** to factors; Convert variables **HomePenalty** and **AwayPenalty** to numeric format and replace the missing values with 0; Select an appropriate chart type to visualize the distribution of **HomePenalty** for matches that involved a penalty shootout. Briefly justify your choice and interpret the resulting plot.

2. Write the code to analyse the distribution of **HomeTeamScore** across different competition stages (**Stage**) and tournament name (**TournamentName**). Then, write the code to investigate the distribution of **HomeTeamScore** across the following groups of **AwayTeamScore** (e.g., 0-1 goals, 2-3 goals, 4 or more goals). Visualise the findings using histograms and briefly explain your observations.

3. Filter out the matches that were decided by a penalty shootout. What are the top 5 national teams that most frequently participated in these matches? Do the rankings differ between home teams and away teams? Provide separate tables containing the columns **TeamName, TeamCode,** and **NumberOfMatches,** showing the top 5 home teams and top 5 away teams involved in penalty shootouts. Briefly discuss your findings.

4. Write the code to analyse the factors that might influence the match outcome (**Result**) in the FIFA World Cup dataset. The **Result** variable consists of three categories: Home Team Win, Away Team Win, and Draw. Any factors in the dataset, such as **Stage**, **ExtraTime,** and **Country**, can be considered. Select two factors and investigate whether there are any trends that explain variations in match outcomes. Support your analysis with appropriate tables and visualisations, and briefly discuss your findings.

5. Create a GitHub repository for this assignment and upload all relevant R scripts and datasets used in the analysis. Make at least three commits during the development process and provide the GitHub repository link in your report.

Write a PDF report containing your code and all required analysis and results. The report is being marked using the marking criteria, so make sure that each piece of analysis covers all of the criteria. The PDF file should also be uploaded to your GitHub repository.